Permutation inference for displaced/non-displaced EEG decoding

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1 Question and fixed primary analysis

Does the fixed eight-channel, 100 ms shrinkage-LDA decoder exceed a participant-specific label-shuffled chance distribution under the existing whole-recording validation? The 23 non-pilot participants form the primary evaluation. mle01 and spe30 informed feature/model selection and remain explicitly exploratory. This note complements the earlier classification results and the participant/session waveform figures.

The primary pipeline is continuous 1–20 Hz BP no-FORCe, with the same accepted trial IDs, event alignment, baseline and quality checks as the saved decoder analysis. Seven 100 ms means per channel over $[0.2, 0.9)$ s provide 56 features. Standardisation and shrinkage LDA (equal class priors, automatic covariance shrinkage) are fitted on training recordings only. No preprocessing or model tuning is repeated or changed here.

2 Permutation method and statistic

We draw 1,999 random label permutations independently within each recording, preserving its D/ND counts. Each permutation refits the models using exactly the existing leave-one-recording-out folds, separately within participant/session. Test predictions are pooled within a session before computing balanced accuracy (mean D/ND recall). Participant statistics average eligible session scores equally; each participant's null statistic uses the same aggregation. Exclusions remain rle13/pre and vco27/pre, whose training folds cannot support the protocol. The analysis covers 73 eligible sessions and 30,210 evaluated trials.

The one-sided Monte Carlo value is

$$p = \frac{1 + \#\{b : T_b \geq T_{\text{obs}}\}}{1 + 1999},$$

with ties included and minimum attainable $p = 0.0005$. Chance_{95} is the empirical 95th percentile of the null (upper order statistic), an unadjusted descriptive threshold. Holm adjustment controls family-wise error across the 23 primary participant tests. Session-level values in the web report are unadjusted exploratory checks, not additional corrected primary claims.

The web report also includes central LDA, the all-eight-channel binned tree, and the Graz-style top-100 amplitude tree. The latter reranks all 8,000 full-epoch candidates using training data inside every permutation and fold. These 69 secondary non-pilot tests form a separate Holm family. The eight pilot/model tests form an exploratory family; correction does not remove bias from earlier pilot model/feature selection. Observed held-out predictions for all four models are checked against the saved analyses.

3 Primary participant results

Eight-channel shrinkage LDA exceeded the within-recording label-shuffle null after Holm correction in 23/23 non-pilot participants. The unadjusted participant-specific 95th-percentile chance thresholds ranged from 52.6% to 54.9%. These thresholds describe the null distribution; adjusted significance is determined by the Holm-corrected p-values, not by crossing the unadjusted threshold alone. Raw primary p-values ranged from 0.0005 to 0.0065; 22/23 participants reached the Monte Carlo minimum of 0.0005. All primary Holm-adjusted p-values were 0.0115. Repeated adjusted values reflect the step-down correction and finite sampling resolution, not identical effect sizes. Values at the Monte Carlo floor do not resolve smaller exact tail probabilities.

Participant	Sessions	Trials	BA (%)	Chance_{95} (%)	p	p_{Holm}
ara13	3	1,325	84.2	53.1	0.0005	0.0115
bal08	3	1,261	87.4	52.9	0.0005	0.0115
bpe06	3	1,140	70.3	53.2	0.0005	0.0115
bro19	3	816	79.2	54.8	0.0005	0.0115
cna07	3	994	84.8	53.5	0.0005	0.0115
cve01	3	1,405	86.6	52.8	0.0005	0.0115
den12	3	1,364	71.1	52.7	0.0005	0.0115
eab31	3	1,423	66.3	52.9	0.0005	0.0115
emo24	3	884	74.4	53.9	0.0005	0.0115
fbo09	3	1,332	82.3	53.0	0.0005	0.0115
gba12	3	1,414	82.6	53.0	0.0005	0.0115
gcr27	3	1,439	73.4	53.3	0.0005	0.0115
lbo12	3	1,265	71.0	53.1	0.0005	0.0115
ldu23	3	1,275	72.6	53.3	0.0005	0.0115
meb17	3	1,010	72.1	53.4	0.0005	0.0115
mka24	3	1,398	77.2	52.9	0.0005	0.0115
nfu22	3	1,435	80.8	52.6	0.0005	0.0115
nsh23	3	1,442	71.8	52.7	0.0005	0.0115
rle13	2	887	86.1	53.6	0.0005	0.0115
sga12	3	1,121	72.5	53.2	0.0005	0.0115
sto20	3	1,291	71.9	53.2	0.0005	0.0115
vco27	2	612	56.8	54.4	0.0065	0.0115
yle21	3	1,112	79.6	54.9	0.0005	0.0115

Table 1: Non-pilot participants. BA is the equal-session mean balanced accuracy. Chance_{95} uses the identical statistic under within-recording label permutations. Holm values refer to the 23-test primary family.

Participant	Sessions	Trials	BA (%)	Chance_{95} (%)	p	p_{Holm}
mle01	3	1,179	75.9	53.0	0.0005	0.0040
spe30	3	1,386	85.8	52.7	0.0005	0.0040

Table 2: Exploratory pilots, shown separately. Their Holm values belong to the eight-test pilot/model family; they are not independent validation of pilot-driven choices.

The secondary model tables and unadjusted session checks are available in the existing DEC web report.

4 Observed performance and chance thresholds

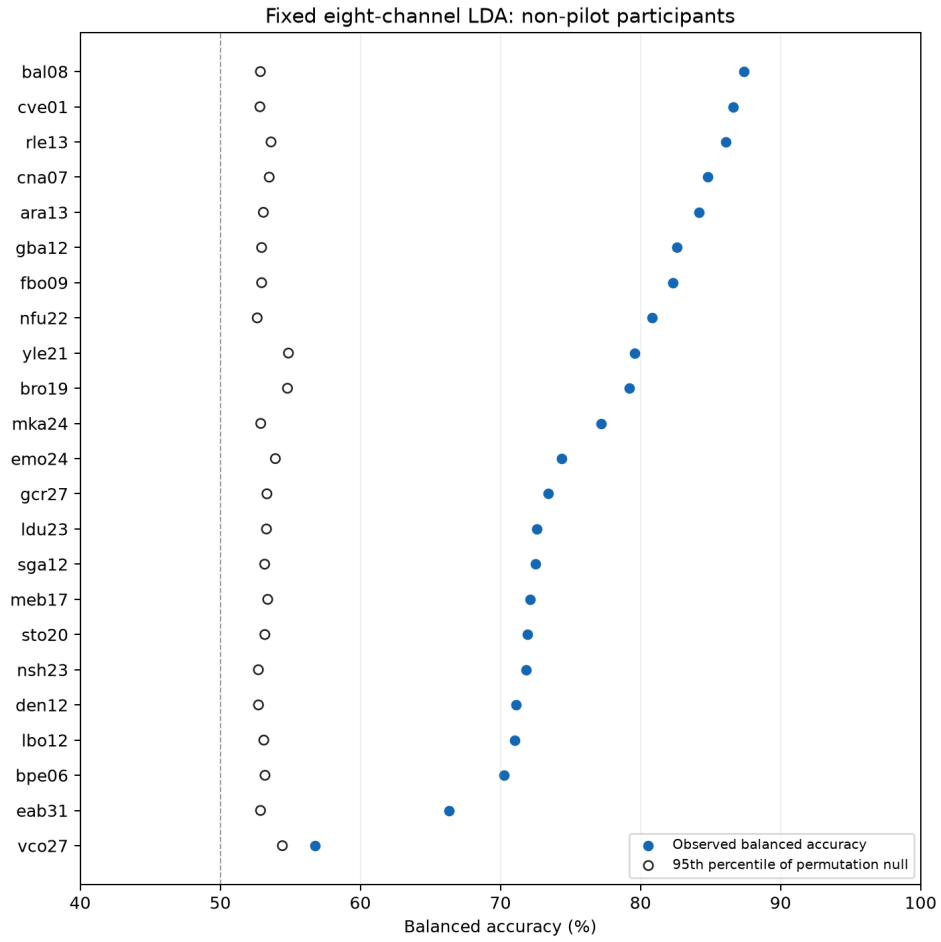


Figure 1: Observed participant scores and unadjusted permutation-derived thresholds for the primary LDA. The threshold markers are not confidence intervals or family-wise corrected decision thresholds.

5 Interpretation and limits

The null assumes D/ND labels are exchangeable within each recording. Blocking by recording preserves recording-level class proportions but does not establish exchangeability in the presence of unmodelled temporal structure. This is inference conditional on the existing trial inclusion and fixed personalised decoder, not a test of generalisation of a trained model to new participants.

Above-null discrimination supports useful label-related information. It does not identify an ErrP, establish error awareness, or rule out visual, movement-related or residual artifact contributions. Waveform morphology, timing and scalp distribution remain complementary evidence.

Method references. Ojala and Garriga, *Permutation tests for studying classifier performance*, JMLR 11 (2010), 1833–1863. Phipson and Smyth, *Permutation p-values should never be zero*, SAGMB 9 (2010), Article 39, doi:10.2202/1544-6115.1585. The implementation follows the grouped-shuffle and +1 Monte Carlo convention documented by scikit-learn.